Jerry Sun

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Technical Skills

Programming: C++, Verilog, Python, JavaScript, MATLAB, VBA
Engineering Tools: Autodesk Fusion, COMSOL, Onshape, Arduino IDE, IBM Maximo
Hardware: Arduino, oscilloscopes, breadboarding, soldering, sensors and motor control
Concepts: PID control, digital logic, embedded systems, asset reliability, predictive maintenance

Experience

Asset Engineer Co-op Feb 2026 – Dec 2026
Capital Power (EPCOR Tower) *Edmonton, Alberta*

- Configured and digitized O&M inspection workflows in IBM Maximo, implementing conditional logic and interactive inspection forms, troubleshooting application errors, and supporting Maximo Mobile implementation.
- Audited and classified extracted metadata for over 175,000 engineering drawings across major power generation facilities using a 61-category technical taxonomy, including P&IDs, single- and three-line diagrams, control diagrams, and structural steel drawings; developed VBA tools to automate validation and standardize outputs.
- Performed QA for a self-learning AI/ML model that scanned facility CAD drawings and generated structured metadata in Excel, reviewing outputs, identifying extraction errors, testing model updates, and providing feedback to improve first-draft accuracy.

Robotics Designer, Builder & Competition Driver Aug 2024 – Now
Team 2055A, UW VEX U Robotics *Oakville/Waterloo, ON*

- Integrated electrical and mechanical systems: wiring motors, sensors, and control modules; diagnosing failures; improving system robustness and maintaining 100% functionality for over 90% of total competition runtime.
- Tuned PID control loops for distance-sensor feedback to improve autonomous positioning accuracy.
- Co-designed and built four competition robots in one season, integrating mechanical systems with programmer-developed control logic. Engineered a piston-actuated clamp and hook intake that scored six 20 cm rings within five seconds at a 95% success rate, along with a passive suspension mechanism that supported the robot without continuous motor power.

Construction Consultant July 2023 – Aug 2023
Sino-Can Energy *Tweed, ON*

- Led a team to construct and assemble over 50 solar panels, collaborating with managers to ensure 100% quality compliance within strict deadlines and maintaining zero project delays over the entire employment duration
- Learned electrical fundamentals hands-on: rectifiers, DC characteristics, wiring safety, mechanical mounting, and system reliability

Projects

Hospital Delirium Heart-Rate Detector | *Embedded Systems, Power & Safety* Oct 2025 – Dec 2025

- Built a heart-rate monitoring prototype to support early delirium detection and safer patient care.
- Prototyped sensor circuits on breadboards, integrating IR components, reducing noise, and stabilizing analog signals and designed an LED-based indication system for immediate recognition of abnormal heart-rate changes.
- Programmed signal-processing logic in C++ (Arduino libraries) to sample data, compute BPM, and trigger real-time alerts.

Prototype Industrial Control & Interlock System | *PLC Basics, Logic* Apr 2026 – Jun 2026

- Built an Arduino-based industrial control prototype with three-wire startstop interlocking logic for momentary push-button inputs.
- Implemented analog signal scaling to convert raw ADC readings into a 0–100% water-level measurement and programmed a high-level safety interlock to shut down the pump actuator above an 85% threshold.

Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science (BASc) in Electrical Engineering — GPA: 3.9/4.0 *Aug 2025 – May 2030*